

10. Spectral Reflectance

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1. Learning Objectives

This module will help us conceptualize the principle of spectral reflectance and understand its significance in studying various earth surface features.

2. Energy Interactions with Earth Surface Features

When electromagnetic energy is incident on any feature of the earth's surface feature, any of the three interactions, viz. reflection, absorption or transmission, may occur. Thus, at any given wavelength for incident energy, the relationship among these interactions can be represented by the following equation (1) based on the principle of conservation of energy:

$$E_i(\lambda) = E_R(\lambda) + E_A(\lambda) + E_T(\lambda) \quad \dots\dots \quad (\text{Equation 1})$$

Where,

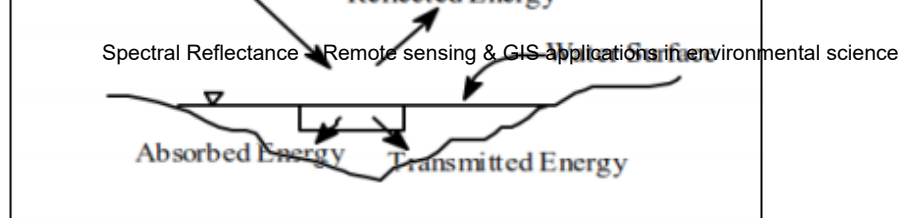


Figure 1: Interactions between electromagnetic energy and Earth surface feature

From equation 1, it becomes imperative to understand that for different earth features, the proportion of energy reflected, absorbed and transmitted varies, depending on the nature of the material. Further, even for a given feature type, the proportion of reflected, absorbed and transmitted energy will vary at different wavelengths.

Since most of the remote sensing systems operate based on the reflected energy, the reflectance properties of the feature become very important for studying earth's surface features. Thus, the energy balance relationship expressed in Equation 1 can be represented as:

$$E_R(\lambda) = E_I(\lambda) - [E_A(\lambda) + E_T(\lambda)] \text{-----(2)}$$

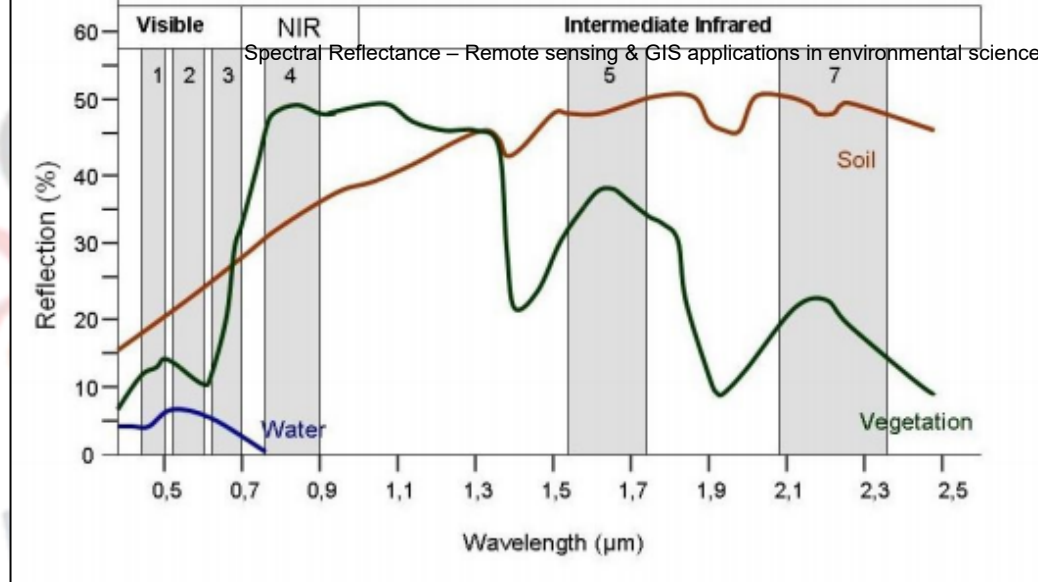
This implies that the reflected energy at any given wavelength is equal to the energy incident on a given feature reduced by the energy that is either absorbed or transmitted by that feature.

termed as spectral reflectance curve. This varies with the variation in the chemical composition and physical conditions of the feature. The spectral response patterns for an object are averaged to get a generalized form, which is called as generalized spectral response pattern. If a spectral response pattern is unique to an object or an earth's feature, it is termed as 'Spectral signature'. Because spectral responses measured by remote sensors over various features often permit an assessment of the type and/or condition of the features, these responses have often been referred to as spectral signatures. However, the term has now become obsolete, and spectral reflectance is only used to characterize a feature. The importance of spectral reflectance curve lies in the fact that it gives us an insight into the spectral characteristics of the object under consideration. This can be illustrated by the examples of spectral reflectance curves of the following earth's surface features.

3.1 Vegetation

All the green plants contain the pigment chlorophyll, which absorbs electromagnetic radiations greatly in the visible region. As a result, for healthy green vegetation, spectral reflectance curve exhibits the "peak-and-valley" configuration (Fig. 2). The peaks indicate predominant reflection and the valleys indicate dip in reflectance due to strong absorption of the energy in the corresponding wavelength bands.

Figure 2: Spectral reflectance curve for vegetation, soil and water
(<http://www.seosproject.eu/modules/remotesensing/remotesensing-c01-p05.html>)



Further, the spectral response of vegetation depends on the structure of the plant leaves. Fig. 2 shows the cell structure of a green leaf and the interaction with the electromagnetic radiation (Gibson 2000). In a plant leaf, the palisade cells containing chlorophyll pigment strongly absorb energy in the wavelength bands centered at 0.45 and 0.67 μm corresponding to blue and red wavelengths within visible region (Fig. 3). This dip in reflectance is known as 'chlorophyll absorption bands'. Also, this reflection is highest for the green colour in the visible region, due to which our eyes perceive healthy vegetation to be green in colour.

As we move from visible region of electromagnetic spectrum to infrared region; at 0.7 μm , the absorption reduces while reflection greatly increases. Then this reflectance is nearly constant from 0.7-1.3 μm where plant leaf reflects about 50 percent of the energy incident upon it. Most of the remaining energy is transmitted, since absorption in the spectral region is minimal (less than 5%). Plant reflectance in the range of 0.7 to 1.3 μm results primarily from the internal structure of plant. The infrared radiation penetrates the palisade cells and reaches the irregularly packed mesophyll cells which make up the body of the leaf. Mesophyll cells reflect almost 60% of the NIR radiation reaching this layer. Most of the remaining energy is transmitted, since absorption in this spectral region is minimal. Healthy vegetation therefore shows brighter response in the NIR region compared to the green region.

Beyond 1.3 μm , energy incident upon the vegetation is essentially absorbed or reflected, with little to no transmittance of energy. Dips in the reflectance occur at the 1.4, 1.9 and 2.7 μm because water in the leaf absorbs strongly at these wavelengths. Accordingly, wavelengths in these spectral regions are referred to as 'water absorption bands'. Reflectance peaks occur at about 1.6 and 2.2 μm , between the absorption bands. Throughout the range beyond 1.3 μm , leaf reflectance is approximately inversely related to the total water present in a leaf. This total is function of both the moisture content and the thickness of a leaf.

Due to high degree of variability between plant species, the significance of reflectance measurements lies in the fact that these measurements help to distin-

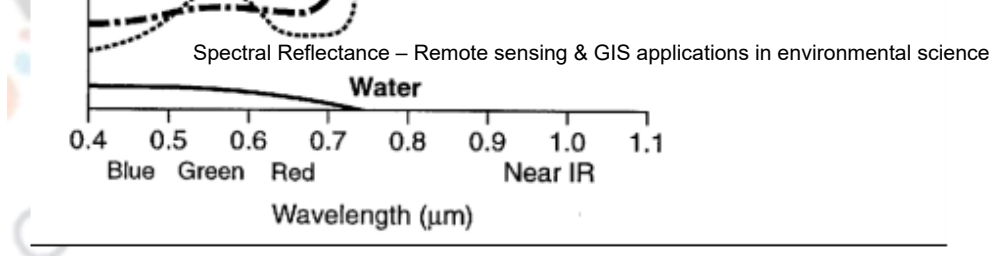


Figure 4: Spectral reflectance curve of various land features

(<https://gis.stackexchange.com/questions/101392/determining-which-color-each-image-band-represents>)

Further, in stressed vegetation, the NIR bands are no longer reflected by the mesophyll cells, instead they are absorbed by the stressed or dead cells causing dark tones in the image. Also, multiple layers of leaves in a plant canopy provide the opportunity for multiple transmittance and reflectance. Hence, the near-IR reflectance increases with the number of layers of leaves in a canopy, with the reflection maximum achieved at about eight leaf layers (Bauer et., 1986).

Another example illustrating the importance of spectral reflectance curve is the application in forestry to distinguish between deciduous versus coniferous trees (Figure 5). It is pertinent to mention that the reflectance curve for these trees is plotted as a ribbon of values, not as a single line to include a broad range of reflectance values, rather than a discrete value for each tree.

From Figure 5, it is clear that if we try to discriminate the two types of trees in visible bands, it would be a bit difficult since the spectral reflectance curves for the two tree types overlap in the visible region. As a result, both the tree types would appear green in colour and would exhibit similar reflectance. Although this difficulty can be overcome partially by using clues such as size and shape of canopy; however, it would not be possible to give absolute results. However, if we have a sensor that can detect the reflectance of the vegetation in infrared bands, this would solve our purpose. Since, deciduous trees have broad leaves compared to needle shaped leaves of conifers; they show much higher reflectance in infrared wavelength and thus, appear much lighter in tone than conifers. Thus, based on spectral characteristics, various Earth surface features can be identified and mapped.

3.2 Soil

The reflectance curve for soil in figure 4 shows considerably less peak and valley variation. The factors that affect soil reflectance are moisture content, organic matter content, soil texture (proportion of sand, silt and clay), surface roughness and presence of iron oxide. The presence of moisture in soil decreases its reflectance due to the presence of water absorption bands at 1.4, 1.9 and 2.7 μm . Clay soil also has hydroxyl absorption bands at 1.4 and 2.2 μm . There also exists close relation with soil moisture content and soil texture; for example, coarse, sandy soils are usually well drained, resulting in low moisture content and relatively high re-

However, various conditions of water bodies manifest themselves primarily in visible wavelengths such as the presence of organic as well as inorganic materials in water. For example, highly turbid water containing large quantities of suspended sediments has much higher visible reflectance. Similarly, the presence of chlorophyll pigment as a result of algal growth decreases the reflectance, a fact used for studying the eutrophication status of water body using remote sensing techniques. Similarly, spectral reflectance is also used for determining the presence of tannin dyes, industrial wastes discharges and various pollutants that alter the reflectance.

3.4 Snow

Snow reflects strongly in the visible and near infrared and absorbs more energy at mid-IR wavelengths that is characterized by a dip in spectral reflectance. However, the reflectance of snow is affected by its grain size, liquid water content and presence or absence of other materials (Dozier and Painter, 2004). Larger grains of snow absorb more energy, particularly at wavelengths longer than 0.8 μm . At temperature near 0°C, liquid water within the snow pack can cause grains to stick together in clusters, thus increasing the effective grain size and decreasing the reflectance at near IR and longer wavelengths. When particles of contaminants such as dust or soot are deposited on snow, they can significantly reduce the surface's reflectance in the visible spectrum.

Figure 6: Spectral reflectance of snow and clouds (<http://www.geol-amu.org/notes/m1r-1-8.htm>)

From Figure 6, it can be inferred that the absorption of mid-IR wavelengths by snow can permit the differentiation between snow and clouds. While both feature types appear bright in the visible and near IR, clouds have significantly higher reflectance than snow at wavelengths longer than 1.4 μm .

3.5 Asphalt

Sand can have a wide variation in its spectral reflectance pattern depending on its parent material.

Besides, presence or absence of water and organic matter also affect the spectral response of sand.

Figure 7: Spectral reflectance curve of various land features

(<http://slideplayer.com/slide/8259758/>)

As shown in the figure 7, the spectral reflectance curves for asphalt is much flatter than those of other land cover features; however, its reflectance may be modified by the presence of paint, soot, or water. Besides, ageing also has an effect on spectral reflectance of asphalt. The reflectance of asphaltic concrete increases on ageing, particularly, in the visible spectrum.

4. Summary

Having studied the spectral reflectance curve of various land features, it would not be wrong to conclude that these features can be separated spectrally based on their properties of absorption, reflectance and transmittance. This spectral variability study can also be affected by temporal and spatial effects. Temporal effects are those factors that change the spectral characteristics of a feature over time. For example, the spectral characteristics of many vegetation species are in a nearly continual state of change throughout a growing season. Spatial effects refer to the factors that cause the same types of features at a given point in time to have different characteristics at different geographic locations. For example, analysing a crop pattern varies if the analysis is carried out on a small scale or a large regional scale where entirely different soils, climates and cultivation practices might exist for the same crop. Another example could be analysing spectral variability for a diseased versus healthy vegetation. However, these effects are extremely helpful

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